Ryo Kitano

548-384-6021 | rkitano@uwaterloo.ca | linkedin.com/in/ryo-kitano | github.com/Ryo0326-hub

EDUCATION

University of Waterloo Waterloo, ON
Bachelor of Mathematics, Honours, Co-operative Program Sep. 2024 – May 2029

- • **Award:** President's Scholarship (2024)
- • **Coursework:** Optimization, Probability, Statistics, Linear Algebra II, Combinatorics, Graph Theory, Network Flow Theory, Algorithm Design & Data Abstraction

EXPERIENCE

Co-Founder & Technical Lead Mar. 2026 – Present
Neural Point Analytica (NPA) *Remote / Japan*

- Co-founded a 3-person software company and lead engineering for an NDA-protected enterprise workflow platform; secured its first recurring client contract worth US$1.5K/month
- Lead architecture and delivery of auditable RFQ intake, quotation, reconciliation, and reporting workflows using Next.js, TypeScript, PostgreSQL, and AWS; turn client feedback into staged releases with Docker and GitHub Actions

Java Backend & Computer Vision Engineer Intern May 2026 – Sep. 2026
AISTGroup LLC *Remote / Baku, Azerbaijan*

- Delivered Spring Boot REST APIs for invoice, receipt, and file workflows in a 5-engineer team, spanning JPA entities, validation, localization, Liquibase migrations, and JUnit tests
- Implemented encrypted SMTP configuration and a transactional outbox that decoupled receipt delivery from API requests and enabled retryable asynchronous processing; hardened behavior through code review and tests
- Owned data preparation, training, and evaluation for a YOLO26-s + EfficientNet-B0 pipeline using 41.9K images and 180K annotated signs; achieved 0.871 mAP@50 detection and 89.0% Top-1 classification across 154 sign classes

PROJECTS

Hybrid Token-Efficient Routing Agent | Python, Qwen, Fireworks, Docker Jul. 2026

- Trained a hashed n-gram logistic router on 360 measured task outcomes to route between quantized Qwen2.5-1.5B and a hosted LLM under a 2-vCPU / 4-GB budget; added deterministic verification and failover
- Passed 80/80 mixed-task evaluations using 9,745 tokens; on a 20-case hard set, optimized escalation prompting raised accuracy from 85% to 95% while cutting token use by 24.6%

KenMemory | Live Site | OpenAI API, FastAPI, React, PostgreSQL/pgvector Mar. 2026 – Aug. 2026

- Independently built and deployed a privacy-aware RAG product; embedded only consented memories in PostgreSQL/pgvector, used HNSW vector search for semantic retrieval, and streamed grounded answers, verified by 63 backend tests

ThetaTrap | Python, Qwen, Alpaca MCP, SQLite Aug. 2026 – Sep. 2026

- Independently built and deployed an MCP-native paper-options agent with deterministic Python controls that locked candidates, strikes, sizing, and max loss before LLM tool use; passed 191 tests and 5/5 mutation-free replays

Kraken Knight | Python, Kraken API, SQLite, systemd Aug. 2026 – Sep. 2026

- Backtested a deterministic BTC/CAD strategy on 4.63M historical Kraken trades, validated it through shadow trading, then deployed live execution with authenticated orders, fill reconciliation, idempotent audit logs, and fail-closed risk controls

TECHNICAL SKILLS

Languages: Python, Java, SQL; C, TypeScript, JavaScript (working knowledge)
AI/ML: Machine/deep learning, PyTorch, scikit-learn, Hugging Face Transformers, LLMs, RAG/vector search, prompt engineering, agentic/multi-agent systems, MCP/tool calling, computer vision, responsible AI, LLM evaluation
Frameworks & Data: TensorFlow/Keras, OpenAI API, FastAPI, React, Next.js, PostgreSQL/pgvector, Pinecone, pandas, NumPy, Matplotlib, Jupyter, REST APIs
MLOps & Developer Tools: Docker, GitHub Actions (CI/CD), AWS, Vercel, Render, Linux, Git/GitHub; Azure, GCP, Kubernetes (working knowledge)